

lab21(SSU)

November 17, 2025

1 Lab 21: Normal Distribution

```
[ ]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install py pandoc
!pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntaxtex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java

```

libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]

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Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
 Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all 2.004.5-6.1 [9,471 kB]
 Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style all 12.2-1ubuntu1 [185 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64 1.41-4build2 [61.3 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64 2.5.11+ds1-1 [699 kB]
 Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-gyre all 20180621-3.1 [6,209 kB]
 Get:42 <http://archive.ubuntu.com/ubuntu> jammy-updates/universe amd64 texlive-binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]

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Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 7s (27.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 121703 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.

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Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...

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Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...

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Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package tlutils.
Preparing to unpack .../38-tlutils_1.41-4build2_amd64.deb ...
Unpacking tlutils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.

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Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...

```

```

Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_openc1.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

```

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running updmmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

This may take some time... done.

Collecting py pandoc

Downloading py pandoc-1.16.2-py3-none-any.whl.metadata (16 kB)

Downloading py pandoc-1.16.2-py3-none-any.whl (19 kB)

Installing collected packages: py pandoc

Successfully installed py pandoc-1.16.2

Collecting datascience

Downloading datascience-0.18.1-py3-none-any.whl.metadata (910 bytes)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from datascience) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from datascience) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datascience) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from datascience) (1.16.3)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from datascience) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from datascience) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from datascience) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages

(from datascience) (0.8.2)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->datascience) (2025.10.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.3.3)

Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (1.4.9)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (25.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience) (2.9.0.post0)

Collecting jedi>=0.16 (from ipython->datascience)

 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)

Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.4.2)

Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.7.5)

Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (5.7.1)

Requirement already satisfied: prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)

Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (2.19.2)

Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.0)

Requirement already satisfied: matplotlib-inline in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.1)

Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (4.9.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-

```

packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.10.5)
Downloading datascience-0.18.1-py3-none-any.whl (725 kB)
725.1/725.1 kB
13.1 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
55.7 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.1 jedi-0.19.2

```

```

[ ]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')

```

Mounted at /content/drive

```
[ ]: # Run this cell, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plots
plots.style.use('fivethirtyeight')
import warnings
warnings.simplefilter('ignore')
```

2 1. Normal Distributions

When we visualize the distribution of a sample, we are often interested in the mean and the standard deviation of the sample (for the rest of this lab, we will abbreviate “standard deviation” as “SD”). These two sample statistics can give us a bird’s eye view of the distribution – by letting us know where the distribution sits on the number line and how spread out it is, respectively.

We want to check if the data is linearly related, so we should look at the data.

First, let’s load in the `births` table from lecture, which is a large random sample of US births and includes information about mother-child pairs.

```
[ ]: births = Table.read_table('./DS/baby.csv')
births
```

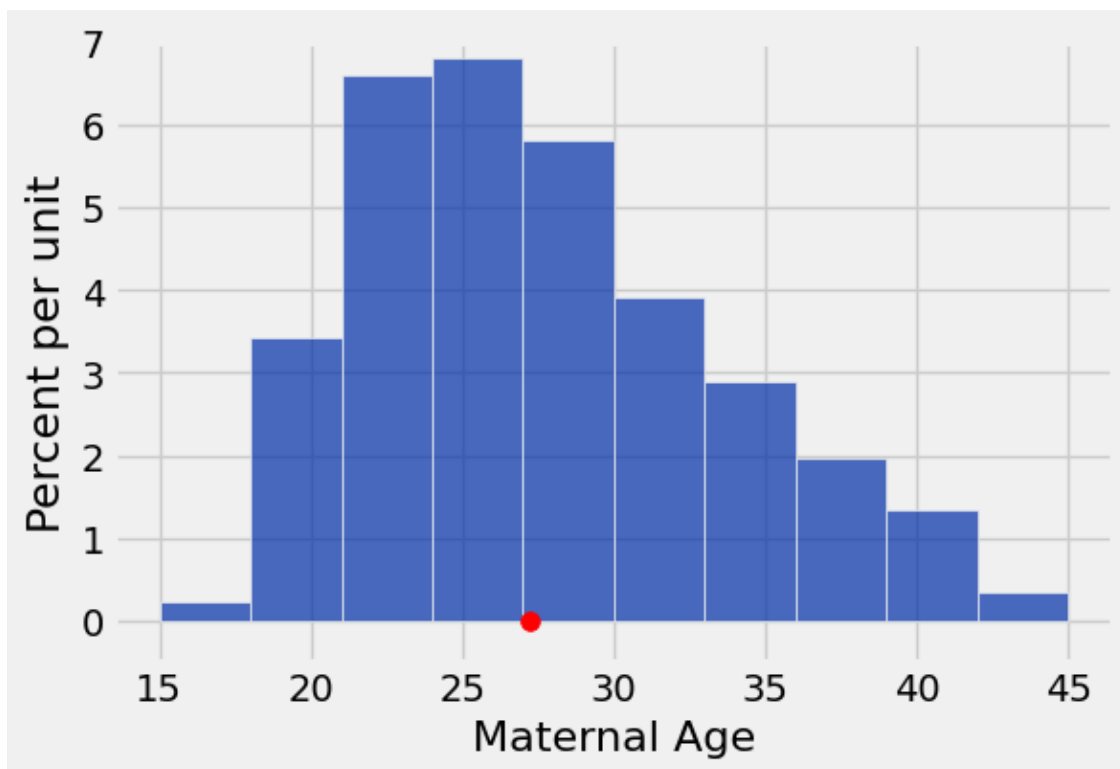
```
[ ]: Birth Weight | Gestational Days | Maternal Age | Maternal Height | Maternal
Pregnancy Weight | Maternal Smoker
120              | 284                | 27              | 62              | 100
| False
113              | 282                | 33              | 64              | 135
| False
128              | 279                | 28              | 64              | 115
| True
108              | 282                | 23              | 67              | 125
| True
136              | 286                | 25              | 62              | 93
| False
138              | 244                | 33              | 62              | 178
| False
132              | 245                | 23              | 65              | 140
| False
120              | 289                | 25              | 62              | 125
| False
143              | 299                | 30              | 66              | 136
```

```
| True
140          | 351          | 27          | 68          | 120
| False
... (1164 rows omitted)
```

Question 1.1. Now, write a line of code to plot the distribution of mother’s ages from the `births` table. Don’t change the last line, which will plot the mean of the sample on the distribution itself.

```
[ ]: births.hist('Maternal Age')

# Do not change anything below this line
plots.scatter(np.mean(births.column("Maternal Age")), 0, color='red', zorder=2,
              ↪s=50);
plots.ylim(-0.005, 0.07);
```

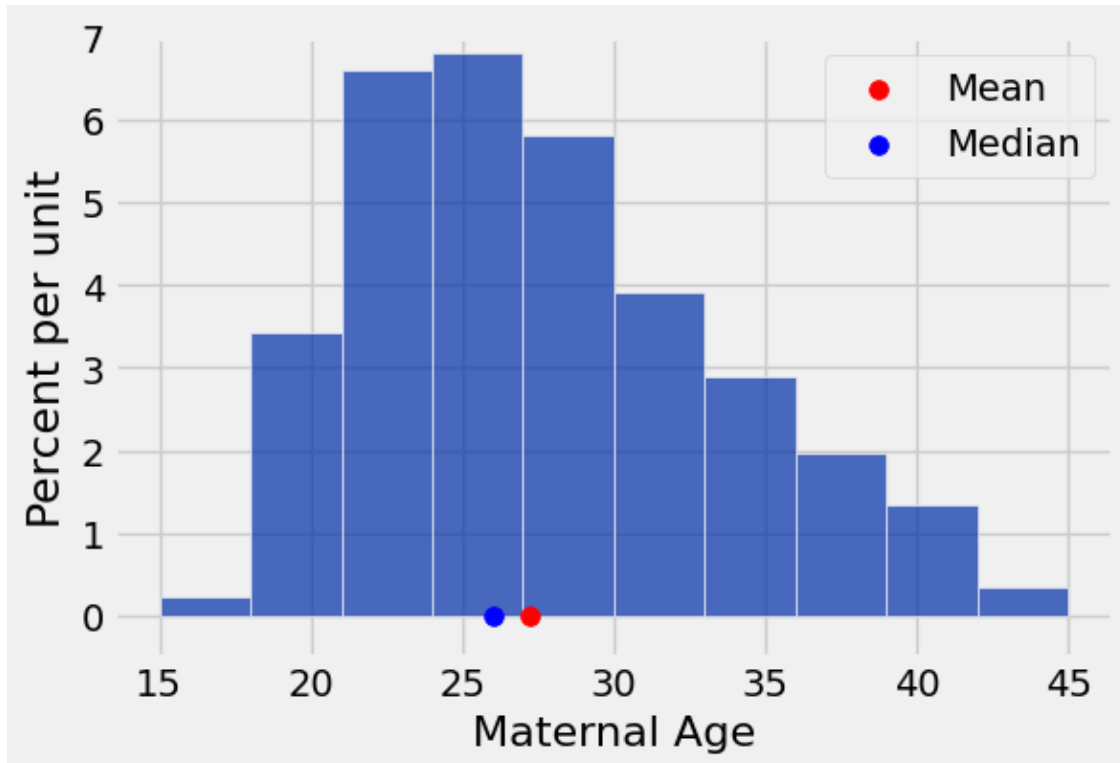


From the plot above, we can see that the mean is the center of gravity or balance point of the distribution. If you cut the distribution out of cardboard, and then placed your finger at the mean, the distribution would perfectly balance on your finger. Since the distribution above is right skewed (which means it has a long right tail), we know that the mean of the distribution is larger than the median, which is the “halfway” point of the data. Conversely, if the distribution had been left skewed, we know the mean would be smaller than the median.

Question 1.2. Run the following cell to compare the mean (red) and median (blue) of the distri-

bution of mothers ages.

```
[ ]: births.hist("Maternal Age")
plots.scatter(np.mean(births.column("Maternal Age")), 0, color='red', zorder=2, s=50, label="Mean")
plots.scatter(np.median(births.column("Maternal Age")), 0, color='blue', zorder=2, s=50, label="Median")
plots.legend()
plots.ylim(-0.005, 0.07);
```



We are also interested in the standard deviation of mother's ages. The SD gives us a sense of how variable mothers' ages are around the average mothers' age. If the SD is large, then the mothers' ages should spread over a large range from the mean. If the SD is small, then the mothers' ages should be tightly clustered around the average mother age.

The SD of an array is defined as the “root mean square of the deviations” where the deviations are the difference between the original elements and their average.

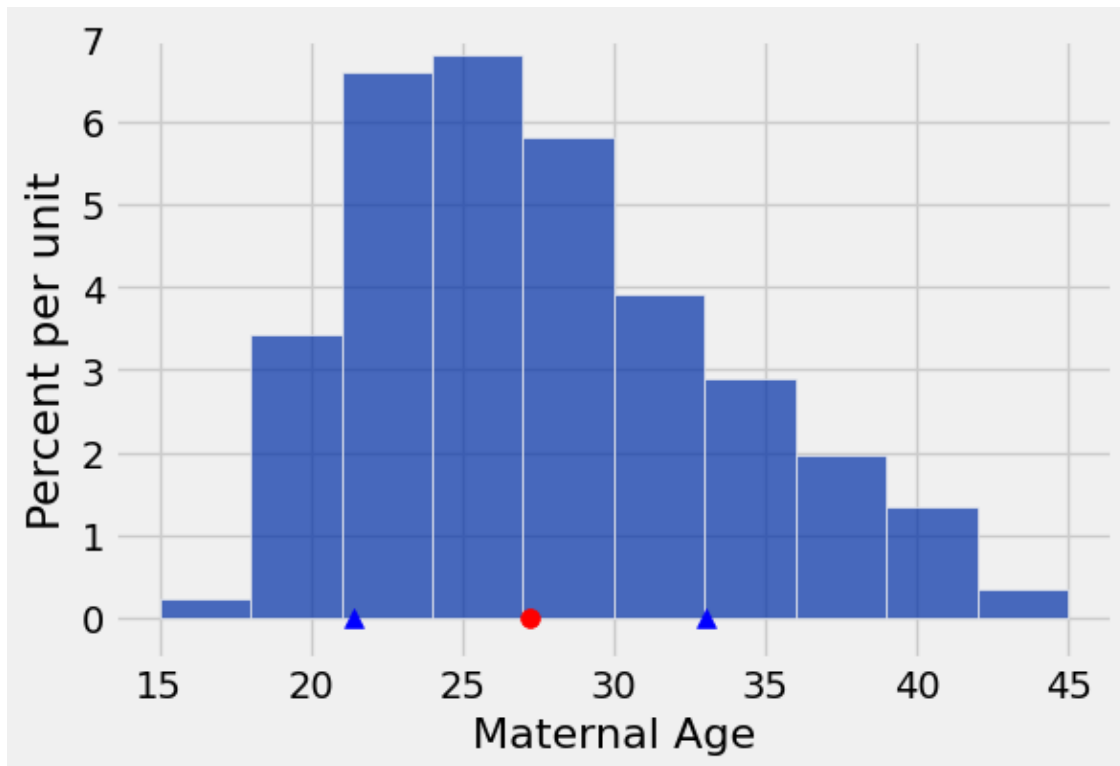
You can read more about Variability and SD in [Chapter 14.2](#). This chapter also goes a bit more into detail about how “root mean square of the deviations” is calculated. Fun fact! σ (Greek letter sigma) is used to represent the SD and μ (Greek letter mu) is used for the mean.

Question 1.3. Assign `age_mean` to the mean of the maternal ages and `age_sd` to the SD of the maternal ages. Then run the cell below to see the width of one SD (blue) from the sample mean (red) plotted on the histogram of maternal ages.

Hint: `np.std` might be useful here.

```
[ ]: age_mean = np.mean(births.column("Maternal Age"))
age_sd = np.std(births.column("Maternal Age"))
births.hist("Maternal Age")

plots.scatter(age_mean, 0, color='red', zorder=2, s=50);
plots.scatter(age_mean+age_sd, 0, marker='^', color='blue', zorder=2, s=50);
plots.scatter(age_mean-age_sd, 0, marker='^', color='blue', zorder=2, s=50);
plots.ylim(-0.005, 0.07);
```



In this histogram, the standard deviation is not easy to identify just by looking at the graph.

However, the distributions of some variables allow us to easily spot the standard deviation on the plot. For example, if a sample follows a **normal distribution**, the standard deviation is easily spotted at the point of inflection (the point where the curve begins to change the direction of its curvature) of the distribution.

Question 1.4. Fill in the following code to examine the distribution of maternal heights, which is roughly normally distributed. We'll plot the standard deviation on the histogram, as before - notice where one standard deviation (blue) away from the mean (red) falls on the plot.

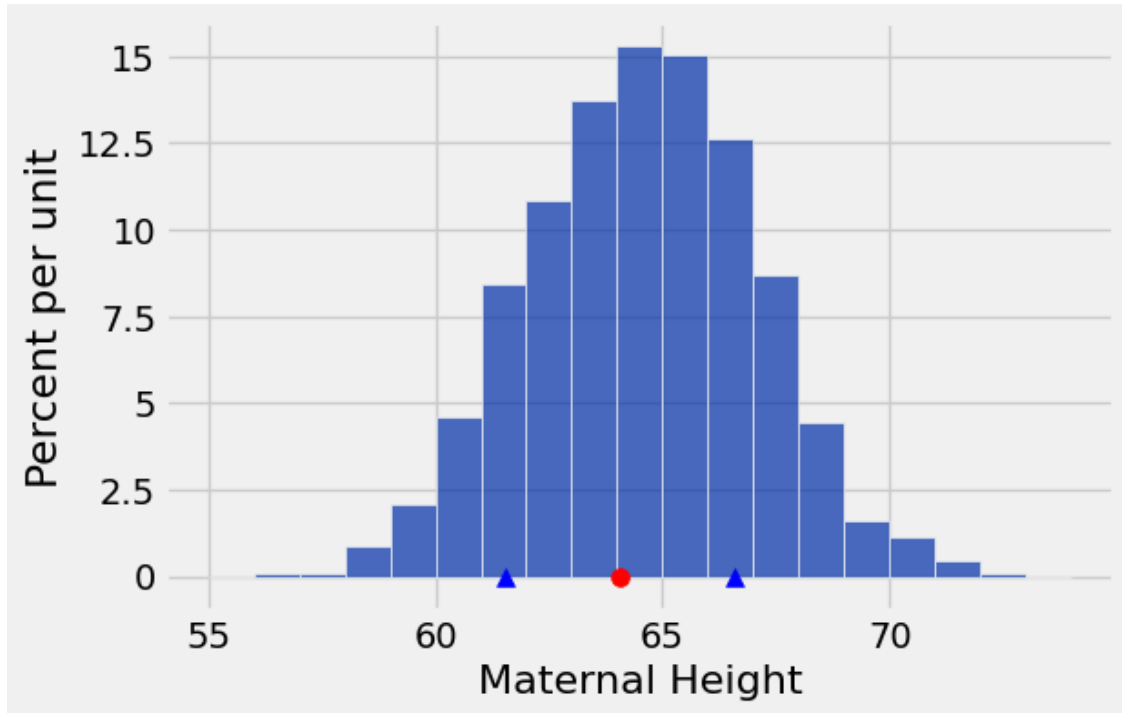
```
[ ]: height_mean = np.mean(births.column("Maternal Height")) #
height_sd = np.std(births.column("Maternal Height")) #
```

```

births.hist("Maternal Height", bins=np.arange(55,75,1)) #
#      (bin) 55~75  1

plots.scatter((height_mean), 0, color='red', zorder=2, s=50);
plots.scatter(height_mean+height_sd, 0, marker='^', color='blue', zorder=2, s=50);
plots.scatter(height_mean-height_sd, 0, marker='^', color='blue', zorder=2, s=50);
plots.ylim(-0.01, 0.16);

```



We don't always know how a variable will be distributed, and making assumptions about whether or not a variable will follow a normal distribution is dangerous. However, the **Central Limit Theorem** defines one distribution that always follows a normal distribution. **The distribution of the *sums* and *means* of all large random samples drawn with replacement (and thus are independent from one another) from a single distribution (regardless of the distribution's original shape) will be normally distributed.** Remember that the Central Limit Theorem refers to the distribution of a *statistic* calculated from a distribution, not the distribution of the original sample or population. If this is confusing, ask a staff member!

2.1 2. Polling and the Normal Distribution

Michelle is a statistical consultant, and she works for a group that supports Proposition 68 (which would mandate labeling of all horizontal and vertical axes, unrelated to any real California proposition) called Yes on 68. They want to know how many Californians will vote for the proposition.

Michelle polls a random sample of all California voters, and she finds that 210 of the 400 sampled voters will vote in favor of the proposition. We have provided a table for you below which has 3 columns: the first two columns are identical to `sample`. The third column contains the proportion of total voters that chose each option.

```
[ ]: sample = Table().with_columns(
    "Vote", make_array("Yes", "No"),
    "Count", make_array(210, 190))

sample_size = sum(sample.column("Count"))
sample_with_proportions = sample.with_column("Proportion", sample.
    ↪column("Count") / sample_size)
sample_with_proportions
```

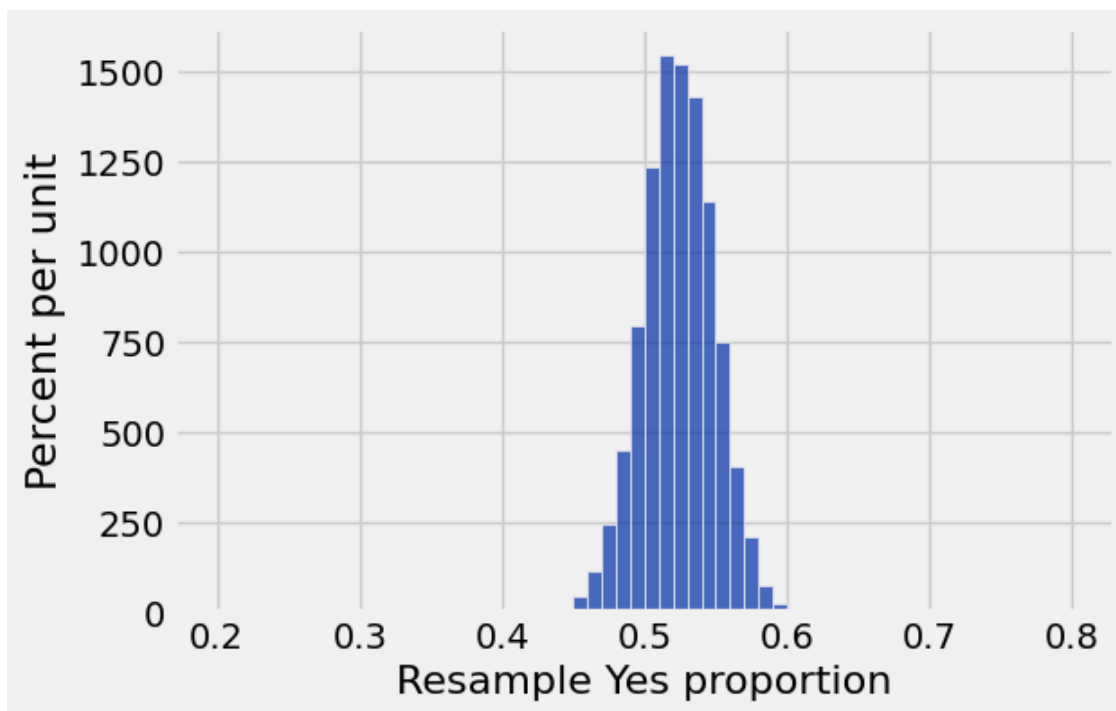
```
[ ]: Vote | Count | Proportion
Yes   | 210   | 0.525
No    | 190   | 0.475
```

Question 2.1. Michelle wants to use 10,000 bootstrap resamples to compute a confidence interval for the proportion of all California voters who will vote Yes.

Fill in the next cell to simulate an empirical distribution of Yes proportions. Use bootstrap resampling to simulate 10,000 election outcomes, and assign `resample_yes_proportions` to contain the Yes proportion of each bootstrap resample. Then, visualize `resample_yes_proportions` with a histogram. **You should see a bell shaped histogram centered near the proportion of Yes in the original sample. (6 points)**

Hint: `sample_proportions` may be useful here!

```
[ ]: resample_yes_proportions = make_array()
for i in np.arange(10000):
    resample = sample_proportions(sample_size, sample_with_proportions.
    ↪column("Proportion"))
    resample_yes_proportions = np.append(resample_yes_proportions, resample.
    ↪item(0))
Table().with_column("Resample Yes proportion", resample_yes_proportions).
    ↪hist(bins=np.arange(.2, .8, .01))
```



Question 2.2. Why does the Central Limit Theorem (CLT) apply in this situation, and how does it explain the distribution we see above? We recommend reviewing [14.4](#) for a refresher on CLT. (6 points)

The Central Limit Theorem (CLT) applies here because we are observing the distribution of a statistic (the sample proportion), not the distribution of individual votes. Since our sample size ($n=400$) is large and our bootstrap samples are drawn randomly, the CLT predicts that the distribution of these sample proportions will be approximately normal (bell-shaped), which is what the histogram shows.

In a population whose members are 0 and 1, there is a simple formula for the **standard deviation of that population**:

$$\text{standard deviation of population} = \sqrt{(\text{proportion of 0s}) \times (\text{proportion of 1s})}$$

(Figuring out this formula, starting from the definition of the standard deviation, is an fun exercise for those who enjoy algebra.)

Question 2.3. Using only the Central Limit Theorem and the numbers of Yes and No voters in our sample of 400, *algebraically* compute the predicted standard deviation of the `resample_yes_proportions` array. Assign this number to `approximate_sd`. **Do not access the data in `resample_yes_proportions` in any way.** (6 points)

Remember that the standard deviation of the sample means can be computed from the population SD and the size of the sample (the formula above might be helpful). If we do not know the

population SD, we can use the sample SD as a reasonable approximation in its place. [This section](#) of the textbook also may be helpful.

```
[ ]: approx_pop_sd = np.sqrt(sample_with_proportions.column("Proportion").item(0) *  
    ↳ sample_with_proportions.column("Proportion").item(1))  
approximate_sd = approx_pop_sd / np.sqrt(sample_size)  
approximate_sd
```

```
[ ]: 0.024968730444297725
```

Question 2.4. Compute the standard deviation of the array `resample_yes_proportions`, which will act as an approximation to the true SD of the possible sample proportions. This will help verify whether your answer to question 2.3 is approximately correct. **(6 points)**

```
[ ]: exact_sd = np.std(resample_yes_proportions)  
exact_sd
```

```
[ ]: 0.024876463595133458
```

Question 2.5. Again, without accessing `resample_yes_proportions` in any way, compute an approximate 95% confidence interval for the proportion of Yes voters in California. **(6 points)**

The cell below draws your interval as a red bar below the histogram of `resample_yes_proportions`; use that to verify that your answer looks right.

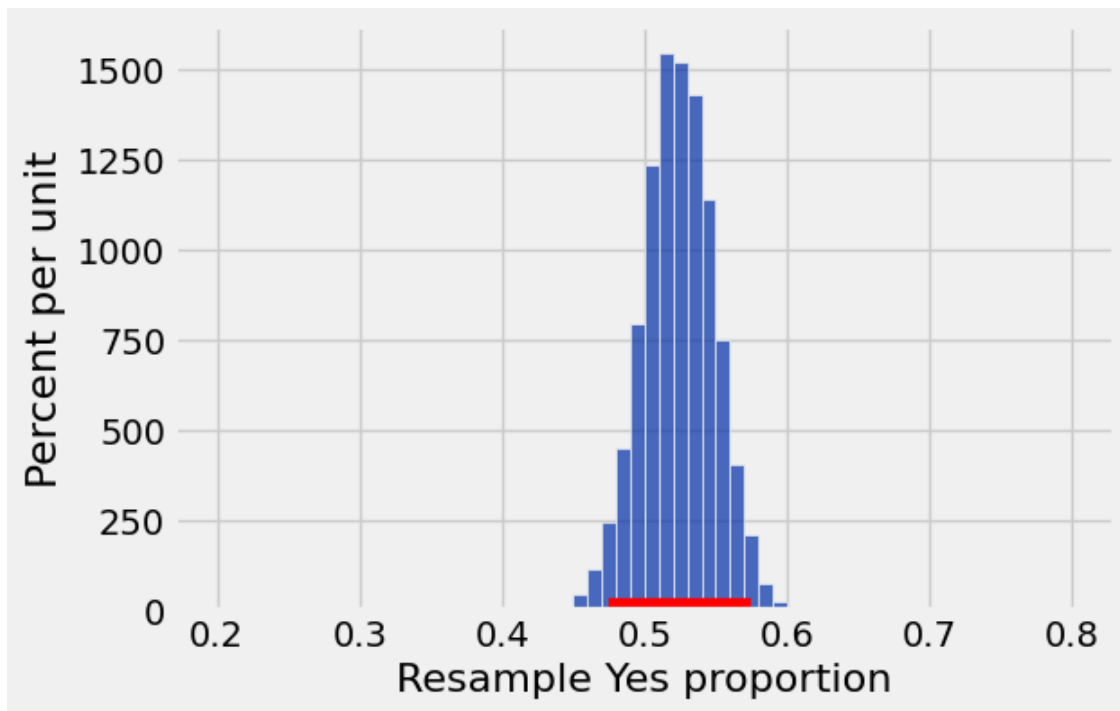
Hint: How many SDs corresponds to 95% of the distribution promised by the CLT? Recall the discussion in the textbook [here](#).

Hint: The `approximate_sd` variable you previously defined may be helpful!

```
[ ]: yes_sample = sample_with_proportions.column("Proportion").item(0)  
lower_limit = yes_sample - 2 * approximate_sd  
upper_limit = yes_sample + 2 * approximate_sd  
print('lower:', lower_limit, 'upper:', upper_limit)
```

```
lower: 0.475062539111 upper: 0.574937460889
```

```
[ ]: # Run this cell to plot your confidence interval.  
Table().with_column("Resample Yes proportion", resample_yes_proportions).  
    ↳ hist(bins=np.arange(.2, .8, .01))  
plots.plot(make_array(lower_limit, upper_limit), make_array(0, 0), c='r',  
    ↳ lw=10);
```



Your confidence interval should overlap the number 0.5. That means we can't be very sure whether Proposition 68 is winning, even though the sample Yes proportion is a bit above 0.5.

The Yes on 68 campaign really needs to know whether they're winning. It's impossible to be absolutely sure without polling the whole population, but they'd be okay if the standard deviation of the sample mean were only 0.005. They ask Michelle to run a new poll with a sample size that's large enough to achieve that. (Polling is expensive, so the sample also shouldn't be bigger than necessary.)

Michelle consults Chapter 14 of the textbook. Instead of making the conservative assumption that the population standard deviation is 0.5 (coding Yes voters as 1 and No voters as 0), she decides to assume that it's equal to the standard deviation of the sample,

$$\sqrt{(\text{Yes proportion in the sample}) \times (\text{No proportion in the sample})}.$$

Under that assumption, Michelle decides that a sample of 9,975 would suffice.

Does Michelle's sample size achieve the desired standard deviation of sample means? What SD would you achieve with a smaller sample size? A higher sample size?

Question 2.6. To explore this, first compute the SD of sample means obtained by using Michelle's sample size and assign it to `michelle_sample_mean_sd`. (6 points)

```
[ ]: estimated_population_sd = np.sqrt(sample_with_proportions.column("Proportion").
    ↪ item(0) * sample_with_proportions.column("Proportion").item(1))
michelle_sample_size = 9975
```

```

michelle_sample_mean_sd = estimated_population_sd / np.
    ↳sqrt(michelle_sample_size)
print("With Michelle's sample size, you would predict a sample mean SD of %f."↳
    ↳% michelle_sample_mean_sd)

```

With Michelle's sample size, you would predict a sample mean SD of 0.005000.

Question 2.7. Next, compute the SD of sample means that you would get from a smaller sample size. Ideally, you should pick a number that is significantly smaller, but any sample size smaller than Michelle's will do. **(5 points)**

```

[ ]: smaller_sample_size = 1000 # Michelle's sample size is 9975, so 1000 is↳
    ↳significantly smaller
smaller_sample_mean_sd = estimated_population_sd / np.sqrt(smaller_sample_size)
print("With this smaller sample size, you would predict a sample mean SD of %f"↳
    ↳% smaller_sample_mean_sd)

```

With this smaller sample size, you would predict a sample mean SD of 0.015792

Question 2.8. Finally, compute the SD of sample means that you would get from a larger sample size. Here, a number that is significantly larger would make any difference more obvious, but any sample size larger than Michelle's will do. **(5 points)**

```

[ ]: larger_sample_size = 50000
larger_sample_mean_sd = estimated_population_sd / np.sqrt(larger_sample_size)
print("With this larger sample size, you would predict a sample mean SD of %f"↳
    ↳% larger_sample_mean_sd)

```

With this larger sample size, you would predict a sample mean SD of 0.002233

Question 2.9. Based off of this, was Michelle's sample size approximately the minimum sufficient sample, given her assumption that the sample SD is the same as the population SD? Assign `min_sufficient` to `True` if 9,975 was indeed approximately the minimum sufficient sample, and `False` if it wasn't. **(4 points)**

```

[ ]: min_sufficient = True
min_sufficient

```

```

[ ]: True

```

2.2 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```

[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab21(SSU).ipynb' --to pdf

```

```
[NbConvertApp] Converting notebook ./DS/lab21(SSU).ipynb to pdf
[NbConvertApp] Support files will be in lab21(SSU)_files/
[NbConvertApp] Making directory ./lab21(SSU)_files
[NbConvertApp] Writing 85349 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 197401 bytes to DS/lab21(SSU).pdf
```